SDC Hackathon 2026

Technical Tools Declaration — Vortex OrderGuard

Instruction compliance: this public copy declares APIs, cloud services, AI tools, datasets, models and sponsor resources used in the project.
Privacy note: leader email/phone and signature are omitted from the public repository copy; complete them in the privately submitted official document.

Team Vortex · Vortex OrderGuard

Prepared 24 October 2026

1. Team & Project Details

Team Name: Vortex

Project / Solution Name: Vortex OrderGuard

Leader: Aun Yi Qi

University: Multimedia University (MMU)

Live Prototype: https://sdc-hackathon-submission-template.replit.app

Code Repository: https://github.com/Aunyiqi/SDC_Hackathon_Submission_Template

Email & Phone: omitted from public repository copy; complete in official private submission.

2.1. System Architecture Overview

Browser UI → Flask web application → optional DeepSeek text extraction and/or RunPod EasyOCR → human-confirmed structured fields → deterministic PO/receiving/invoice discrepancy engine → Trust Gate + Supplier Pattern Memory → owner approval/rejection → simulated outbox + audit history → SQLite demo storage.

AI and OCR only extract or structure evidence. They do not calculate final business outcomes, infer unit conversions, blacklist suppliers, approve actions or send real supplier messages.

2.2. Languages, Frameworks & Libraries

Name & Version	Purpose	License
Python 3.13	Backend language	PSF License
Flask 3.1.3	Web framework	BSD-3-Clause
Requests 2.34.2	HTTP client	Apache-2.0
Gunicorn 26.2.0	Production WSGI server	MIT
RunPod SDK 1.12.0	Serverless worker/API SDK	MIT
EasyOCR 1.7.2	OCR extraction	Apache-2.0
OpenCV / Pillow / NumPy	OCR/image dependencies	Upstream open-source licenses
HTML / CSS / JavaScript	Frontend	Team code / web standards

3.1. External APIs

API and Provider	Purpose	Access Tier
DeepSeek API (optional)	Supplier-message structured extraction; manual fallback when unavailable	Team API account / provider pricing
RunPod Serverless API	Authenticated invocation of GPU OCR worker	Sponsor credit / team account
No live Orbyte API	Orbyte integration is simulated/aligned for MVP	None

3.2. Cloud Platforms & Deployment

Service & Provider	Purpose	Plan / Region
Replit	Public HTTPS web app hosting and publication	Team Replit account; public deployment
RunPod Serverless	GPU compute for EasyOCR inference	Sponsor credit / serverless GPU
GitHub	Public repository and GitHub Actions CI	Public repository / hosted CI

4.1. AI Models Used in the Solution

Model & Version	Source / Provider	Usage	License / Terms
EasyOCR 1.7.2 pre-trained OCR models	JaidedAI / EasyOCR	Pre-trained OCR inference on RunPod GPU	Apache-2.0
deepseek-chat (optional API model)	DeepSeek	Pre-trained hosted model for supplier text extraction	DeepSeek provider terms

4.2. Datasets

Dataset Name	Source / URL	Type	License	Personal Data?
OrderGuard synthetic demo records	Team-created in application	Synthetic	Team-created	No
Synthetic invoice sample	docs/sample-invoice.png	Synthetic	Team-created	No
No training dataset	N/A	None — no model training/fine-tuning performed	N/A	No

4.3. AI-Assisted Development Tools

Did the team use generative AI for development? Yes.

Tool & Version	Used For
ChatGPT / OpenAI	Ideation, architecture review, code/document drafting, debugging assistance, pitch/submission material
Replit Agent	Deployment and Replit configuration troubleshooting
DeepSeek (where used during development/testing)	Idea generation / optional application extraction integration

4.4. Review & Responsibility for AI-Generated Output

The team reviewed generated suggestions before using them, ran automated tests for core matching/security behavior, manually verified deployment and RunPod evidence, and retained human approval for consequential workflow actions. AI-generated code or text was adapted to the project rather than treated as authoritative. Known limitations are documented, including optional provider dependencies and OCR extraction failures. The team remains responsible for all submitted code and claims.

5. Official SDC Sponsor-Provided Resources

Sponsor	Resource	How It Was Used
RunPod	US$25 sponsor credit/voucher issued for team use	GPU serverless OCR proof-of-use with EasyOCR; successful RTX 5090 inference recorded

5.1. Is the Sponsor Resource Essential?

RunPod GPU compute demonstrates the OCR acceleration path but is not essential to the core OrderGuard discrepancy engine. The application can continue with manual structured input if OCR is unavailable. The OCR layer could be replaced by another document-extraction service or CPU-based workflow, while the deterministic comparison, Trust Gate and owner approval logic remain unchanged.

6.1. Data Handling & Privacy

The public demo is intended for synthetic records only. No real customer personal data is required. Synthetic supplier messages or invoice images may be sent to the configured DeepSeek or RunPod service only when those optional features are used. API keys and owner access codes are stored as server-side environment secrets and are not committed to the repository. Demo cases and audit events are stored in SQLite until the demo is explicitly reset. Provider-side request retention follows the provider account/service settings; therefore only synthetic data should be submitted.

6.2. Known Limitations & Risks

OCR can misread text or return incomplete fields; a captured RunPod test completed on GPU but returned empty structured invoice fields, so OCR remains optional and requires human confirmation. Hosted AI output may be inaccurate, so deterministic rules—not the model—calculate discrepancies. The MVP uses SQLite and process-local throttling, not production multi-tenant infrastructure. There is no live Orbyte API integration. Direct SME user validation and measured productivity impact are still pending. Mitigations include manual fallback, Trust Gate blocking, owner-only actions, synthetic data, audit history and automated regression tests.

7. Submission Checklist

• [x] APIs, cloud services, AI tools, models and datasets used by the project are declared.

• [x] Third-party libraries/models are listed with license/terms information where known.

• [x] No API keys, passwords or credentials are included.

• [x] Generative-AI development assistance is disclosed.

• [x] RunPod sponsor credit is declared.

• [x] Public live prototype and repository links are provided.

• [ ] Public GitHub copy intentionally omits leader email/phone and handwritten signature; complete those in the official private submission.

8. Team Leader Confirmation

By submitting the final private copy, the team leader confirms that all declared resources are complete and accurate to the best of the team’s knowledge.

Signature: Omitted from public GitHub copy.

Date: 24 October 2026